

Lab 8 Dipole Moment of Polar Molecules in Solution 5/2

1. Literature Values

From the CRC handbook, σ -Dichlorobenzene:

$$M = 147.002 \text{ g/mol} \quad \rho = 1.3059 \text{ g/cm}^3$$

$$n_D = 1.5515 \quad \mu = 2.50 \pm 0.05 \text{ D}$$

2. Linear Regressions

* Regression 1 (K v.s. X_2^{lit}): $K = K_1^0 + aX_2$ (eq. 20) $\Rightarrow \begin{cases} \text{intercept} = 2.0396 \pm 0.0047 \\ \text{slope} = 6.2832 \pm 0.1698 \end{cases}$

$$\Rightarrow a = 6.2832 \pm 0.1698; K_1^0 = 2.0396 \pm 0.0047$$

* Regression 2 (ρ v.s. X_2^{lit}): $\rho = \rho_1^0 + bX_2$ (eq. 21) $\Rightarrow \begin{cases} \text{intercept} = 0.7750 \pm 0.0004 \\ \text{slope} = 0.5725 \pm 0.0147 \end{cases}$

$$\Rightarrow b = 0.5725 \pm 0.0147; \rho_1^0 = 0.7750 \pm 0.0004$$

* Regression 3 (n_D^2 v.s. X_2^{lit}): $n^2 = (n_1^0)^2 + cX_2$ (eq. 27) $\Rightarrow \begin{cases} \text{intercept} = 2.0354 \pm 0.0002 \\ \text{slope} = 0.3651 \pm 0.0076 \end{cases}$

$$\Rightarrow c = 0.3651 \pm 0.0076; (n_1^0)^2 = 2.0354 \pm 0.0002$$

* Room Temperature

$$\bar{T} = \frac{T_1 + T_2 + T_3 + T_4}{4} = \frac{19.4 + 19.6 + 19.7 + 19.8}{4} ^\circ\text{C} = 19.6250^\circ\text{C} = 292.7750 \text{ K}$$

$$\sigma_T = \sqrt{\frac{1}{4-1} \sum_{i=1}^4 (T_i - \bar{T})^2} = \sqrt{\frac{1}{3} (0.225^2 + 0.225^2 + 0.225^2 + 0.175^2)} \approx 0.1708$$

$$\bar{\sigma}_T = \frac{\sigma_T}{\sqrt{N}} = 0.0854$$

$$\lambda_T = \frac{6T}{w} \cdot t = \frac{0.1708}{2} \cdot 3.1824 = 0.2718 \text{ K}$$

$$So, T = 292.7750 \pm 0.2718 \text{ K}$$

3. Hedstrand Method

(i) Calculations

$$\textcircled{1} \text{ Eq. 22: } P_2^0 = \frac{3a}{(K_1^0+2)^2} \cdot \frac{M_1}{P_1^0} + \frac{K_1^0-1}{K_1^0+2} \left[\frac{M_2}{P_1^0} - \frac{M_1 b}{(P_1^0)^2} \right]$$

$$\text{Given } K_1^0 = 2.0396, a = 62832, P_1^0 = 0.7750, b = 0.5725, M_1 = 84.1620,$$

$$M_2 = 147.0020, P_2^0 = \frac{3 \cdot 62832}{4.0396^2} \cdot \frac{84.1620}{0.7750} + \frac{1.0396}{4.0396} \left(\frac{147.0020}{0.7750} - \frac{84.1620 \cdot 0.5725}{0.7750^2} \right)$$

$$\approx 153.6106 \text{ cm}^3 \text{ mol}^{-1}$$

$$\textcircled{2} \text{ Eq. 23: } P_{2,d}^0 = \frac{(n_2^0)^2 - 1}{(n_2^0)^2 + 2} \cdot \frac{M_2}{P_2^0}$$

$$\text{Given } n_2^0 = 1.5515, M_2 = 147.0020 \text{ g} \cdot \text{mol}^{-1}, P_2^0 = 1.3059 \text{ g} \cdot \text{cm}^{-3}$$

$$P_{2,d}^0 = \frac{1.5515^2 - 1}{1.5515^2 + 2} \cdot \frac{147.0020}{1.3059} \approx 35.9415 \text{ cm}^3 \text{ mol}^{-1}$$

$$\textcircled{3} \text{ Eq. 24: } P_{2,\mu}^0 = P_2^0 - P_{2,d}^0 = \frac{1}{3\epsilon_0 N_A} \cdot \frac{\mu^2}{3kT}$$

$$P_{2,\mu}^0 = 153.6106 - 35.9415 = 117.6691 \text{ cm}^3 \text{ mol}^{-1} = 1.1767 \times 10^{-4} \text{ m}^3 \text{ mol}^{-1}$$

$$\mu = \sqrt{\frac{9\epsilon_0 kT P_{2,\mu}^0}{N_A}} = \sqrt{\frac{9 \cdot 8.8542 \times 10^{-12} \cdot 1.3806 \times 10^{-23} \cdot 292.7750 \cdot 1.1767 \times 10^{-4}}{6.0221 \times 10^{23}}} \text{ C} \cdot \text{m}$$

$$\approx 7.9334 \times 10^{-30} \text{ C} \cdot \text{m}$$

$$(1D = 3.3356 \times 10^{-30} \text{ C} \cdot \text{m}) \approx 2.3784 D$$

2) Error Analysis

1) Error of P_2^0

$$\sigma_{P_2^0} = \sqrt{\left(\frac{\partial P_2^0}{\partial a} \sigma_a\right)^2 + \left(\frac{\partial P_2^0}{\partial b} \sigma_b\right)^2}$$

$$\frac{\partial P_2^0}{\partial a} = \frac{3}{K_1^0 + 1} \cdot \frac{M_1}{P_1^0} = \frac{3}{4.03962} \cdot \frac{84.1620}{0.7750}$$

$$\frac{\partial P_2^0}{\partial b} = -\frac{K_1^0 - 1}{K_1^0 + 2} \cdot \frac{M_1}{(P_1^0)^2} = \frac{1.0396}{4.0396} \cdot \frac{84.1620}{(0.7750)^2}$$

From linear regression: $\sigma_a = 0.1698$, $\sigma_b = 0.0147$

$$\sigma_{P_2^0} = \sqrt{\left(\frac{3}{4.03962} \cdot \frac{84.1620}{0.7750} \cdot 0.1698\right)^2 + \left(\frac{1.0396}{4.0396} \cdot \frac{84.1620}{(0.7750)^2} \cdot 0.0147\right)^2}$$

$$\approx 3.4312$$

$$\text{So, } P_2^0 = (153.6106 \pm 3.4312) \text{ cm}^3 \text{ mol}^{-1}$$

Propagate to $P_{2,\mu}$: $P_{2,\mu} = (1.1767 \pm 0.0343) \times 10^4 \text{ m}^3 \text{ mol}^{-1}$

2) Error Analysis of μ

$$\sigma_\mu = \frac{1}{2} \mu \sqrt{\left(\frac{\sigma_{P_{2,\mu}}}{P_{2,\mu}}\right)^2 + \left(\frac{\sigma_T}{T}\right)^2} = \frac{1}{2} \cdot 2.3784 \cdot \sqrt{\left(\frac{3.4312}{117.6641}\right)^2 + \left(\frac{0.0854}{292.15}\right)^2}$$

$$\approx 0.0347$$

$$\text{So, } \mu = (2.3784 \pm 0.0347) \text{ J/K}$$

4. Smith-Guggenheim Method

(1) Calculations

$$\textcircled{1} \text{ Eq. 28: } P_{2,\mu}^0 = \frac{3M_1}{P_1^0} \cdot \left[\frac{a}{(K_1^0 + 2)^2} - \frac{C}{[(K_1^0)^2 + 2]^2} \right]$$

$$\text{Given } M_1 = 84.1620, P_1^0 = 0.7750, a = 1.2832, K_1^0 = 2.0396, C = 0.3651 \cdot (11.0)^2 = 2.154$$

$$P_{2,\mu}^0 = \frac{3 \cdot 84.1620}{0.7750} \cdot \left[\frac{1.2832}{4.0396^2} - \frac{0.3651}{(2.0396^2 + 2)^2} \right] \approx 1/8.1368 \text{ cm}^3 \cdot \text{mol}^{-1}$$

$$= 1.1814 \times 10^{-4} \text{ m}^3 \cdot \text{mol}^{-1}$$

$$\textcircled{2} \text{ Eq. 24: } P_{2,\mu}^0 = P_2^0 - P_{2,d}^0 = \frac{1}{3\epsilon_0 N_A} \cdot \frac{\mu^2}{3kT}$$

$$\mu = \sqrt{\frac{9\epsilon_0 k T P_{2,\mu}^0}{N_A}} = \sqrt{\frac{9 \cdot 8.8542 \times 10^{-12} \cdot 1.3806 \times 10^{-23} \cdot 297.750 \cdot 1.1814 \times 10^{-4}}{6.0221 \times 10^{23}}} \text{ C} \cdot \text{m}$$

$$\approx 7.9492 \times 10^{-30} \text{ C} \cdot \text{m}$$

$$(1D = 3.3356 \times 10^{-30} \text{ C} \cdot \text{m}) \approx 2.3831 D$$

(2) Error Analysis

① Error of P_2^0

$$\sigma_{P_2^0} = \sqrt{\left(\frac{\partial P_2^0}{\partial a} \sigma_a \right)^2 + \left(\frac{\partial P_2^0}{\partial C} \sigma_C \right)^2}$$

$$\frac{\partial P_2^0}{\partial a} = \frac{3}{(K_1^0 + 2)^2} \cdot \frac{M_1}{P_1^0} = \frac{3}{4.0396^2} \cdot \frac{84.1620}{0.7750}$$

$$\frac{\partial P_2^0}{\partial C} = -\frac{3M_1}{P_1^0 [(K_1^0)^2 + 2]^2} = -\frac{3 \cdot 84.1620}{0.7750 (2.0396^2 + 2)^2}$$

$$\text{From linear regression: } \sigma_a = 0.1698, \sigma_C = 0.0076$$

$$\sigma_{P_2^0} = \sqrt{\left(\frac{3}{4.0396^2} \cdot \frac{84.1620}{0.7750} \cdot 0.1698 \right)^2 + \left(\frac{3 \cdot 84.1620}{0.7750 (2.0396^2 + 2)^2} \cdot 0.0076 \right)^2} \approx 3.3934 \text{ cm}^3 \cdot \text{mol}^{-1}$$

$$\text{So, } P_{2,\mu} = (1.1814 \pm 0.0339) \times 10^{-4} \text{ m}^3 \cdot \text{mol}^{-1}$$

② Error Analysis of μ

$$\delta\mu = \frac{1}{2}\mu \sqrt{\left(\frac{\delta P_{2,\mu}}{P_{2,\mu}}\right)^2 + \left(\frac{\delta T}{T}\right)^2} = \frac{1}{2} \cdot 2.3831 \cdot \sqrt{\left(\frac{3.3934}{118.1368}\right)^2 + \left(\frac{0.0854}{292.75}\right)^2}$$

$$\approx 0.0342$$

$$\text{So, } \mu = (2.3831 \pm 0.0342)$$

5. Error Rate

$$\text{Literature: } \mu^{\text{lit}} = (2.50 \pm 0.05)$$

(1) Hedestrand Method

$$\text{Error rate} = \frac{|2.3831 - 2.50|}{2.50} \times 100\% \approx 4.864\%$$

(2) Smith-Guggenheim Method

$$\text{Error rate} = \frac{|2.3831 - 2.50|}{2.50} \times 100\% \approx 4.671\%$$